

Shitao Jin

PhD Candidate in Architecture | AI-Enabled Architectural Programming & Megaproject Decision-Making

PhD candidate in Architecture at Tongji University and joint-training researcher at The University of Melbourne. My work links architectural programming, construction/project management, and AI-enabled decision-making for megaprojects, transportation hubs, child-friendly communities, and learning environments.

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Google Scholar | LinkedIn

289

Google Scholar citations

8

h-index

21

publications / papers

146

verified peer reviews

2

PhD national
scholarships

7.0

IELTS

RESEARCH PROFILE

- Built a coherent research program around architectural programming as the front-end decision system of complex public projects.
- Developed AI-enabled and multi-criteria decision-making approaches for megaproject programming, risk assessment, complexity measurement, stakeholder analysis, and public concern analysis.
- Extended architectural programming research from building briefing into urban community renewal, child-friendly environments, transportation hubs, and social infrastructure.
- Architectural programming / briefing for complex public projects
- AI-enabled decision-making, collective intelligence, and multi-agent reinforcement learning

EDUCATION & ADVISORS

SEP 2025 - PRESENT

Joint PhD Training / Visiting Graduate Researcher, The University of Melbourne

Faculty of Engineering and Information Technology, Department of Infrastructure Engineering. Supervisors: A/Prof Felix Kin Peng Hui; A/Prof Paulo Vaz Serra.

SEP 2023 - JUN 2027 (EXPECTED)

Ph.D. in Architecture, Tongji University

College of Architecture and Urban Planning, Department of Architecture. GPA: 3.60/4.00. Supervisor: Prof. Huijun Tu.

SEP 2020 - JUN 2023

Master of Architecture, Huazhong University of Science and Technology

GPA: 3.48/4.00.

SEP 2015 - JUN 2020

Bachelor of Architecture, Jilin Jianzhu University

SELECTED PUBLICATIONS

Shitao Jin; Huijun Tu; Jiangfeng Li; Yuwei Fang; Zhang Qu; Fan Xu; Kun Liu; Yiquan Lin. Enhancing Architectural Education through Artificial Intelligence: A Case Study of an AI-Assisted Architectural Programming and Design Course. [75 cites]

Buildings, 14(6), 1613. DOI: 10.3390/buildings14061613

Shitao Jin; Lei Peng. Classroom perception in higher education: The impact of spatial factors on student satisfaction in lecture versus active learning classrooms. [52 cites] Frontiers in Psychology, 13, 941285. DOI: 10.3389/fpsyg.2022.941285

Shitao Jin. Current status and trends of megaproject research: bibliometric and text mining analysis. [8 cites]

Engineering, Construction and Architectural Management, 33(6), 4543-4573. DOI: 10.1108/ECAM-04-2024-0495

Shitao Jin. Measuring complexity in mega construction projects: fuzzy comprehensive evaluation and grey relational analysis. [7 cites]

Engineering, Construction and Architectural Management, 33(1), 284-317. DOI: 10.1108/ECAM-07-2024-0951

Shitao Jin. Interdisciplinary perspective on architectural programming: current status and future directions. [14 cites]

Engineering, Construction and Architectural Management, 32(11), 7332-7372. DOI: 10.1108/ECAM-04-2024-0408

Huijun Tu; Shitao Jin. A group decision-making model for architectural programming in megaprojects. [11 cites]

Engineering, Construction and Architectural Management, 31(13), 342-368. DOI: 10.1108/ECAM-03-2024-0394

CORE EVIDENCE

- National Scholarship for PhD Students (2024, 2025); National Scholarship for Master Students (2022)
- 146 verified WOS/Publons reviews across 7 journals
- Joint training at The University of Melbourne in infrastructure engineering, technology management, and construction management

METHODS

- Python, machine learning, natural language processing, deep learning, multi-agent reinforcement learning
- Bibliometrics, LDA topic modeling, sentiment analysis, social network analysis, entropy weight, fuzzy TOPSIS, Monte Carlo simulation
- Rhino, Grasshopper, Revit, AutoCAD, SketchUp, Photoshop, Illustrator, InDesign, Premiere Pro

TEACHING FIT

- AI-assisted architectural programming/design course model published in Buildings (2024)
- Can teach: architectural programming, AI for design, project/construction management, research methods

ACADEMIC SERVICE

- 146 verified reviews during 2024-2026; main reviewer for ECAM, Buildings, and Sustainability
- Class monitor during master's study; academic committee and employment leadership as undergraduate

PRACTICE

JUL 2022 - AUG 2022

Architectural Management Intern

China Railway Siyuan Survey and Design Group Co., Ltd.

JUL 2021 - SEP 2021

Architectural Design Intern

China Architecture Design & Research Group

JUN 2019 - AUG 2019

Architectural Design Intern

Wuhan Hechuang Architectural Design Co., Ltd.